

Perishable Rescue + Equity Coordinator

Five specialized agents replace the manual decision bottleneck at a food bank —
safety-first, equity-aware, and it learns every run.

Deterministic + LLM · Deployable today · rescue.intrudr.io

60M

Pounds of food handled every year,
across **400+ partner agencies** and **510
weekly pickups**.

95%

Of modern rescued food is **highly
perishable** — fresh produce, protein
and dairy, not shelf-stable cans.

TIN

The problem isn't a lack of food —
sorting and routing are simple for
fresh goods.

A 48-hour clock the manual desk can't beat

| PHYSICAL LIMITS

Not enough warehouse cold storage to hold perishables · a severe shortage of refrigerated transport trucks.

| MANUAL PROCESSING

No automated flagging for lots about to expire · one operator juggles many complex variables under time pressure.

0 h ————— 48 h

Berries and leafy greens deteriorate in **hours**. Raw chicken and meat are safe for only **1–2 days**. Without fast processing, fresh food spoils waiting on the dock.

▶ TWO FAILURES, ONE ROOT CAUSE

| OPERATIONAL LOSS

5-7%

of **already-rescued** food is lost — food that was paid for and delivered.

≈ 3-4 million lbs wasted annually · roughly 2-3 million meals discarded.

| EQUITY FAILURE

1 in 9

182,080 Alameda County residents are food insecure (11%).

East Oakland & Fruitvale show severe need — hunger concentration traces decades-old redlining maps.

▶ THE SILENT EQUITY LEAK · DEMAND DRIFT



KEY TAKEAWAY — It's a decision bottleneck, not a food shortage. A standing order is a **flawed signal**; following it blindly locks in the wrong food mix.

| STATUS QUO

One monolithic manual decision

A single operator juggles spoilage, equity, trucks *and* communication at once. Complex variables blur together — and one of them gets **silently dropped** under pressure.

| OUR DESIGN

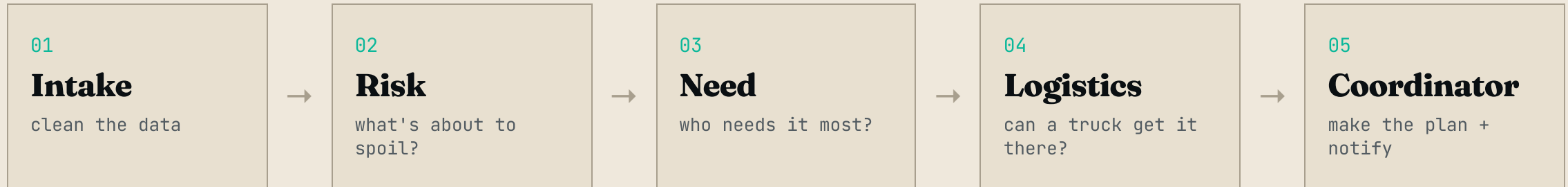
Five agents, one constraint each

Facts are handled by **deterministic code**; judgment by **LLM reasoning**. Strict boundaries mean an AI opinion **never blurs with a physical safety rule** — nothing gets silently ignored.

► FACTS VS JUDGMENT · THE SPLIT THAT MATTERS

	DETERMINISTIC – MATH & PHYSICS	LLM – CONTEXT & EQUITY
ROLE	Strictly calculates reality and enforces safety.	Weighs competing human variables.
MECHANISM	Plain code – the exact same answer every time.	AI reasoning inside structured prompt guardrails.
TASKS	Spoilage math (hours-until-unsafe) · truck capacity · route feasibility.	Ranking who needs it most · correcting demand drift · multilingual notices.
RULE	Physics wins. Safety is never reasoned away.	"Who needs it most" requires real judgment.

► THE FIVE-AGENT PIPELINE



| 🧠 THE LEARNING LAYER

**Sharper
every run**

- Risk & Need agents **read past lessons before deciding.**
- The system **records new patterns** (e.g. a recurring demand drift) after every run.
- **Never crashes** — every step has a rule-based fallback. Run #1 is good; run #100 is exceptional.

Agent 1 · Intake no AI

INPUT

Messy raw CSVs — inventory, shelf-life, routes.

ACTION

Parses unstructured records into clean, workable data structures.

OUTPUT

Quantified lots & total pounds — 20 lots · 8,630 lbs.

Agent 2 · Risk deterministic math

ACTION

Computes $(\text{effective shelf-life} - \text{age}) \times 24$.

OUTPUT

Exact **hours-until-unsafe** for every lot.

OUTCOME

Auto-flags critical lots (raw meat at 0 h) with a plain-English urgency tier — an LLM explains the edge cases.

Agent 3 · Need LLM

Applies a **drift boost** to correct agencies under-ordering nutritious food, then outputs a ranked 1-through-5 agency list for every lot.

Ranks who needs it most →

Agent 4 · Logistics hard gate

Checks cold-chain limits, depletes truck capacity sequentially, and cascades to runner-up agencies — issuing a **hard veto** on any unsafe match.

Physics wins over vibes →

Agent 5 · Coordinator LLM

Publishes the final **Rescue Rate** and writes pickup notices in guaranteed **English, Spanish & Chinese**.

Writes the multilingual plan

► PROVEN AGAINST THE STATUS QUO – COMPUTED EVERY RUN

METRIC	STATUS QUO (MANUAL DESK)	MULTI-AGENT PIPELINE
Rescue rate	71%	92% +21 pts
Spoiled on warm trucks (cold-chain ignored)	1,100 lbs	0 lbs – flagged, not lost
Delivered to highest-need agencies (need ≥ 75)	0 lbs	2,600 lbs

The status quo honors orders as written and dispatches the nearest truck – so it spoils cold food on warm routes and sends nutrition to whoever already orders it, leaving **0 lbs** for the neighborhoods furthest behind. Status quo = "honor standing orders · nearest truck · no equity or cold-chain gating".

92%

Same-day rescue rate

3,450 of 3,750 at-risk pounds placed immediately.

DRIFT CORRECTED

Steered **2,600 lbs** of fresh produce & protein to high-need East and West Oakland.

CAPACITY CASCADED

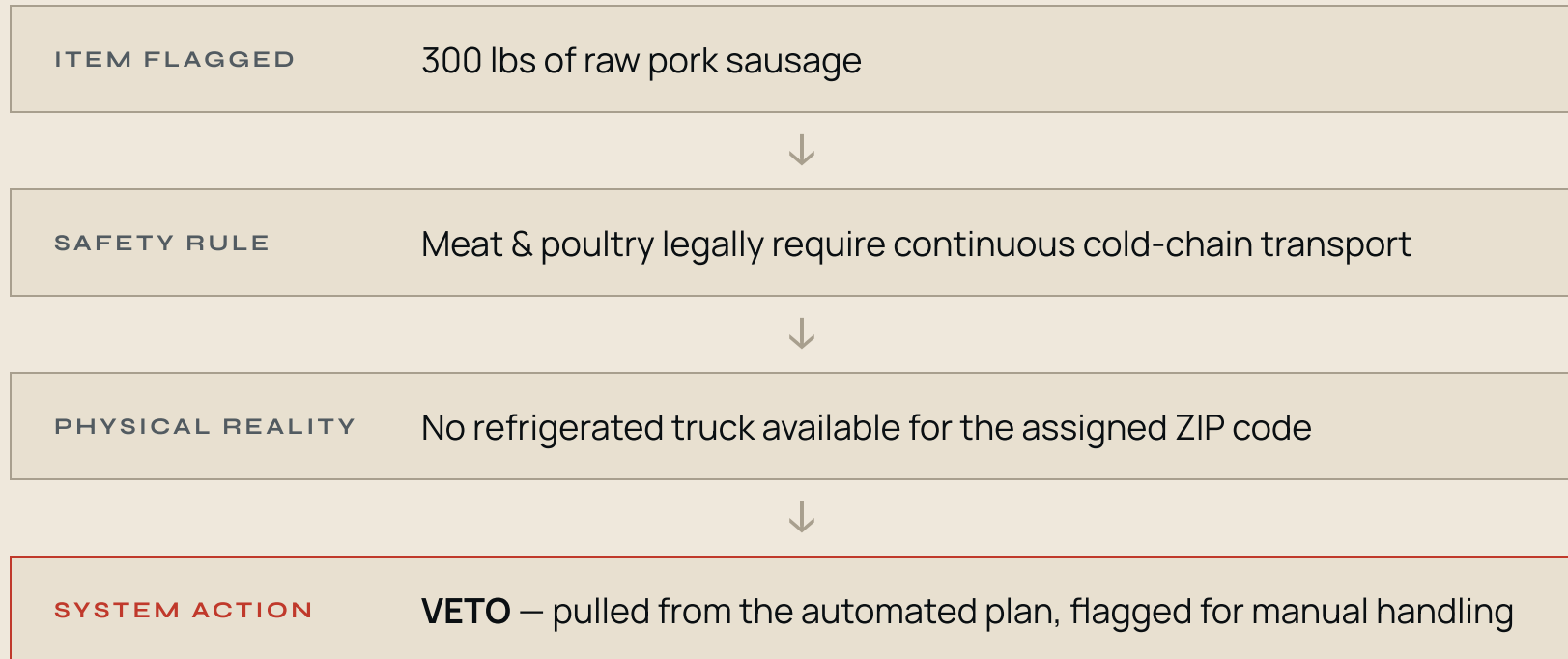
Rerouted **850 lbs** of poultry to Hayward when primary targets ran out of intake.

CONTINUOUS LEARNING

Logged spoilage patterns and recorded A01/A03 demand drift for tomorrow's run.

▶ SAFETY OVERRULES OPTIMIZATION · THE 300 LB VETO

A perfect 100% rescue rate would signal a **dangerous, unconstrained AI**. The 92% outcome proves the system respects absolute safety bounds.



CONCLUSION — Physics wins over vibes. Safety overrules optimization.

Built to plug into ACCFB's real systems, not rewritten for a demo

LIVE IN PRODUCTION

Running at rescue.intrudr.io on a real VPS – auto-deploys on every push.

NEVER FAILS ON THE MODEL

Every LLM step degrades to a rule-based fallback – **identical plan with no API**.

CHEAP & MODULAR

~\$0.001 per run · clean module boundaries so each agent can point at their live inventory / CRM.

PILOT-READY & LOW-RISK

Advisory, not autonomous. Run each morning, compare to the coordinator's plan, measure the rescue-rate delta.

▶ RUN IT LIVE

See the five agents execute in real time



rescue.intrudr.io

Press **Run pipeline** — animated data flow, live agent reasoning, the status-quo contrast, and the final rescue plan.

► OPEN SOURCE · JUDGED DELIVERABLE

The whole build is on GitHub

github.com/Mossab28/perishable-rescue

Five agents · LangGraph orchestrator · learning layer · live
SSE demo · real USDA + Alameda data. README,
architecture & data sources included.

92% rescue rate · +21 pts vs status quo · deployable